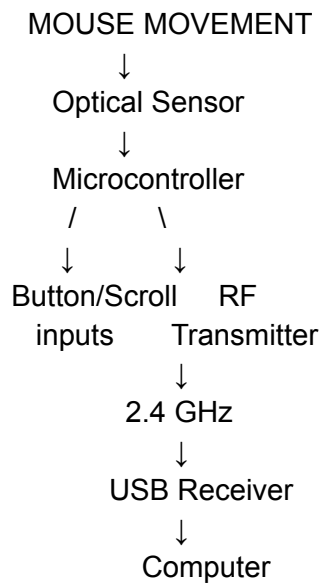


How does a Mouse know when you move it? || How Does a Computer Mouse Work?

First, understand the whole PCB

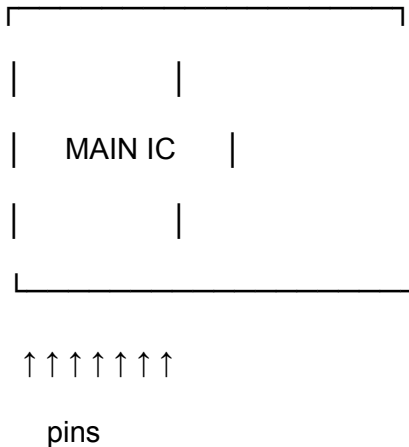
The basic architecture is approximately:



1 Main IC — the black chip in the center

This is the most important visible component.

You can see the black IC around the center:



It appears to be a **multi-function mouse controller IC** or a controller associated with the wireless mouse electronics.

It may contain some combination of:

- Microcontroller
- Mouse protocol logic
- RF interface
- GPIO
- Power-management functions
- Clock interface

The exact chip number in your photograph isn't clear enough for me to confidently identify the manufacturer/model.

What does it do?

Imagine you move your mouse:

Move mouse →



Sensor detects ΔX , ΔY



Controller receives data



Controller creates mouse packet



Wireless transmission

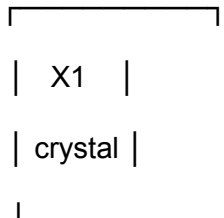


PC

So this chip is essentially one of the **brains of the mouse**.

2 The small metal component marked X1

Near the IC you can see a small metallic rectangular component:



This is likely a **crystal oscillator**.

Why does the mouse need a crystal?

Digital electronics need a clock.

For example:

Crystal



Clock signal



MCU/RF



synchronized operation

The crystal provides a highly stable frequency reference.

Without a stable clock, communication timing could become unreliable.

For wireless communication, accurate timing is particularly important.

3 The tiny resistors marked R...

You can see components labelled something like:

R8

R?

R?

These are **resistors**.

Resistors are used for things such as:

- Current limiting
- Voltage division
- Pull-up/pull-down signals
- Signal conditioning
- Setting circuit parameters

For example:

VCC

|

[R]

|

LED

|

GND

The resistor prevents excessive current from flowing through the LED.

4 Capacitors — C...

You'll also see tiny components marked around the PCB with **C** designators.

For example:

C3

C?

C?

These are **capacitors**.

One of their most important jobs in a mouse is **power-supply filtering**.

Imagine the battery provides:

Battery

↓

VCC ————— IC

The voltage isn't perfectly clean.

A capacitor helps suppress rapid voltage fluctuations:

Battery

↓

VCC ———— IC

|

C

|
GND

This is extremely important for digital and RF circuits.

5 The large screw/metal-looking object

Near the top-right area of your PCB you can see a large circular metal-looking part.

That is **probably not an electronic component**.

It appears to be part of the **mechanical mounting/assembly of the mouse**.

In other words, don't confuse every metal object inside the mouse with an electronic component.

6. The black plastic structure on the right

The large black plastic pieces are mechanical parts of the mouse.

They are likely associated with the:

- Mouse buttons
- Scroll mechanism
- Internal mechanical structure

The PCB sits underneath/around these mechanical components.

So:

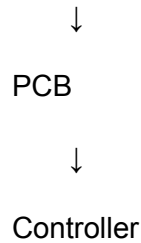
Plastic mechanism



Physical button movement



Electrical switch/contact



7. The vertical row of soldered pins

This is particularly interesting.

On the right side of the IC you can see a **vertical row of solder joints/pins**.

Something like:

-
-
-
-
-
-
-
-

These are electrical connections between PCB traces and another component/assembly.

Depending on the exact mouse design, such connections can carry:

- VCC
- GND
- Data
- Clock
- Button signals
- Sensor signals
- Other control signals

We shouldn't assume every one is a separate signal without tracing the PCB.

8. The copper tracks

Look closely at the green PCB.

You can see thin lines running everywhere:

IC ————— resistor
|
|————— capacitor
|
L————— connector

These are **copper traces**.

They work like wires.

Instead of using individual wires:

IC ——— wire ——— resistor

the PCB creates the electrical connection directly:

IC ===== resistor

copper

trace

This is basically the **nervous system of the PCB**.

9. GND — Ground

Your PCB clearly has a point labelled:

GND

Ground is extremely important.

For example:

Battery +

|

|— IC

|

|— circuit

|

Battery -

|

GND

GND provides the common electrical reference and return path.

10. VDD / VCC

You can also see power-related markings around the PCB.

Something like:

VDD

This generally indicates a positive supply rail for the IC/circuit.

So conceptually:

Battery

|

|— VDD/VCC → electronics

1

```
└─ GND    → return
```

At the bottom of your picture, you have a **flat flexible cable/wire assembly**.

that carries electrical signals between this PCB and another part of the mouse.

Depending on the exact design, it could connect to:

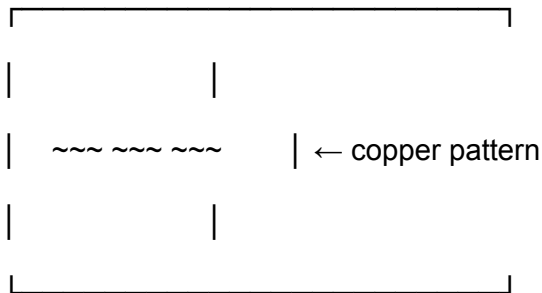
- Optical sensor assembly
- Buttons
- Scroll encoder
- Another PCB
- Power/charging circuitry

We would need to see the other end to identify it with confidence.

12

This is the really interesting part.

can see a **meandering copper trace** near the PCB edge:



That type of PCB trace can be part of an **antenna**, but from this single photo I wouldn't conclusively label every visible zig-zag trace as the RF antenna.

A 2.4-GHz wireless mouse needs an antenna to transmit data:

Controller



RF circuit



Antenna



~ ~ ~ ~ ~

2.4 GHz

~ ~ ~ ~ ~



USB receiver



Computer

So what happens when YOU move this mouse?

Let's connect everything together.

You move the mouse:

YOUR HAND



Mouse moves



Desk texture



Optical image sensor



$\Delta X, \Delta Y$



Main controller



Mouse packet



RF circuit



ANTENNA



2.4 GHz 



USB receiver



COMPUTER



CURSOR 

And simultaneously:

Click button



Mechanical switch



Controller



Wireless packet



PC



Mouse click

And:

Scroll wheel



Encoder



Controller



Wireless packet



PC



Scroll

